

Database Design for a Real-World Application

Application: Easy Shopping – Online Shopping Platform

1. Name and Purpose of the Application

Easy shopping is an online shopping platform that allows customers to browse products, place orders, make payments and track deliveries. The database stores customer, product, order, payment and delivery information in a structured way.

2. List of Required Tables

- Customers
- Categories
- Products
- Orders
- Order items
- Payments
- Deliveries

Seven tables are used. Each table represents an important part of the shopping and order process.

3. Columns / Attributes for Each Table

Customers

Column	Type	Notes
customer_id	INT	Primary key
name	VARCHAR(100)	

phone	VARCHAR(15)	
email	VARCHAR(100)	
address	VARCHAR(255)	Delivery address

Categories

Column	Type	Notes
category_id	INT	Primary key
category_name	VARCHAR(100)	

Products

Column	Type	Notes
product_id	INT	Primary key
category_id	INT	Foreign key -> Categories
product_name	VARCHAR(100)	
price	DECIMAL(10,2)	
stock	INT	Available quantity

Orders

Column	Type	Notes
order_id	INT	Primary key
customer_id	INT	Foreign key -> Customers
order_date	DATETIME	
status	VARCHAR(20)	placed / shipped / delivered / cancelled
total_amount	DECIMAL(10,2)	

OrderItems

Column	Type	Notes
order_item_id	INT	Primary key
order_id	INT	Foreign key -> Orders
product_id	INT	Foreign key -> Products
quantity	INT	
subtotal	DECIMAL(10,2)	

Payments

Column	Type	Notes
payment_id	INT	Primary key
order_id	INT	Foreign key -> Orders
amount	DECIMAL(10,2)	
payment_mode	VARCHAR(20)	UPI / card / cash
payment_status	VARCHAR(20)	pending / completed / failed

Deliveries

Column	Type	Notes
delivery_id	INT	Primary key
order_id	INT	Foreign key -> Orders
delivery_address	VARCHAR(255)	
delivery_date	DATE	
delivery_status	VARCHAR(20)	pending / shipped / delivered

4. Relationships Between the Tables

- Customers -> Orders: One-to-Many (one customer can place many orders)
- Categories -> Products: One-to-Many (one category can contain many products)

- Orders -> OrderItems: One-to-Many (one order can contain many items)
- Products -> OrderItems: One-to-Many (one product can appear in many orders)
- Orders -> Payments: One-to-One (each order has one payment record)
- Orders -> Deliveries: One-to-One (each order has one delivery record)

5. Purpose of Each Table

- **Customers** - Stores customer account and contact details.
- **Categories** - Stores product groups such as electronics, clothing and books.
- **Products** - Stores the products, prices and available stock.
- **Orders** - Stores the main order details and order status.
- **OrderItems** - Stores the products and quantities included in an order.
- **Payments** - Records payment amount, mode and status.
- **Deliveries** - Stores delivery address, date and delivery status.

6. Conclusion

The ShopEasy database uses seven related tables to manage customers, products, orders, payments and deliveries. Primary keys identify records and foreign keys connect related tables.